Experimental Evaluation of SO-100 Positional Accuracy Using Optical Tracking

Group A2

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Final Report

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ME – 360 Engineering Experimentation

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Abstract

Low-cost open-source robot arms such as the LeRobot/Hugging Face SO-100 provide an accessible platform for education and research, with a leader–follower kit costing less than $230, but their motion-reproduction accuracy is not well characterized. This experiment evaluated the 3D end-effector deviation of an SO-100 follower arm relative to a manually operated SO-100 leader arm using a Vicon optical motion-capture system and reflective markers. The initial setup attempted multi-joint tracking with 40 markers across two arms, but tracking loss and coordinate inversion made the joint-level data unreliable. Analysis focused on a simplified end-effector tracking configuration. The Vicon position data was processed in Python. Across 88 static hold positions, the follower end-effector had a mean 3D leader–follower error of 21.895 ± 1.256 mm, a median error of 21.132 ± 1.723 mm, and a standard deviation of 11.963 ± 1.143 mm. These results indicate that the SO-100 follower can reproduce the general trajectory of the leader arm, but with centimeter-scale end-effector deviation, it is unsuitable for tasks requiring millimeter-level positional accuracy without additional calibration, compensation, or mechanical improvements.

Introduction

Motivation

Advancements in physical intelligence have increased the use of robotic systems in research and education, but high-performance robot arms are still very expensive. This can often discourage hobbyists, students, and institutions from using them. Cheap open-source alternatives are now widely available and are increasingly being used in these applications, but their repeatability and precision are unknown. The LeRobot-Hugging Face SO-100 robotic arms are a family of open-source robotic arms in this category, designed to reduce the cost barrier by using 3D-printed components and inexpensive hardware. A complete kit for building a leader and follower arm pair costs less than $230 USD. This lower cost makes the SO-100 platform accessible to a larger audience.

Our client, Professor Eric Lima, utilizes the LeRobot Hugging Face SO-100 robotic arms for research at the Cooper Union. Due to the unknown level of precision these arms offer, a proposal was made to characterize the precision of the SO-100 robotic arm.

Objective

The objective of this project is to design and implement an experimental system that quantifies the repeatability, precision, and trajectory deviation of the SO-100 robotic arm platform. The experimental system must operate within realistic constraints, including repeatable trials with other follower arms and accurate measurements with less than 1 mm in deviation.

Design Constrains and Technical Requirements

The measurement resource chosen for this experimental system was the Vicon Vantage optical motion-capture system. This motion capture system uses infrared cameras and reflective tracking markers to capture three-dimensional motion at high data rates, making it suitable for measuring position deviations between the master and follower robotic arm. The selected measurement approach uses IR reflector balls mounted on the robotic arm links so that the position of each link segment can be measured.

The specific problem addressed in this experiment is how accurately and consistently the SO-100 follower arms can reproduce the motion of an SO-100 master arm via the teleoperation mode. This step is critical as it ensures that good data can be fed to the physical intelligence model.

The process is as follows: a human operator moves the master arm through an arbitrary trajectory which is mirrored by a follower arm. The mocap system is used to record the physical motion of the master and follower arms. The resulting trajectories were compared to quantify follower deviation from the master, precision of motion reproduction across trials, and variability between different follower arms.

Experimental Design

Apparatus

The experiment used a parallel tracking layout consisting of one master SO-100 robotic arm and one SO-100 follower arms at a fixed distance from each other. The master arm was operated manually by a human user. The motor's angular position values for this motion were encoded and sent to a computer connected to the arms. This was then sent to the follower arm, which attempted to reproduce the same motion, with some time delay between the input and output not relevant for this experiment, as final position deviations were of interest. All arms were placed within the optical motion-capture area so that their physical motion could be measured directly.

Each robotic arm has 6 degrees of freedom: shoulder pan, shoulder lift, elbow flex, wrist flex, wrist roll, and the gripper. These degrees of freedom are visualized in Figure 1. All degrees of freedom were measured, except for the gripper section due to mechanical issues with the sample robots that were provided by the client for the purposes of the experiment. The baseplate of each robot was also made as a measurement point to serve as a local coordinate origin to which joint movements could be referenced from.

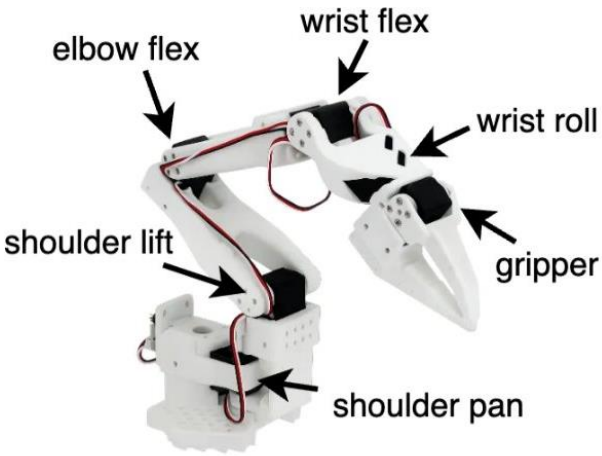


Figure 1: Degrees of Freedom of a SO-100 Robotic Follower Arm

To track the motion of each arm link, four IR reflector balls were attached to each link or degree of freedom of the arm. The placement of four markers allowed the motion-capture system to determine both position and angular orientation of the tracked members relative to the global coordinate system. For easier attachment and repeatable measurement across multiple sample robots, special clips were developed to mount onto each link without causing interference in motion. The clips were designed to index onto reference faces on each link to allow for the repeatable and precise measurement of multiple sample robots. The design and layout of the clips used on the follower arms is shown on Figure 2.

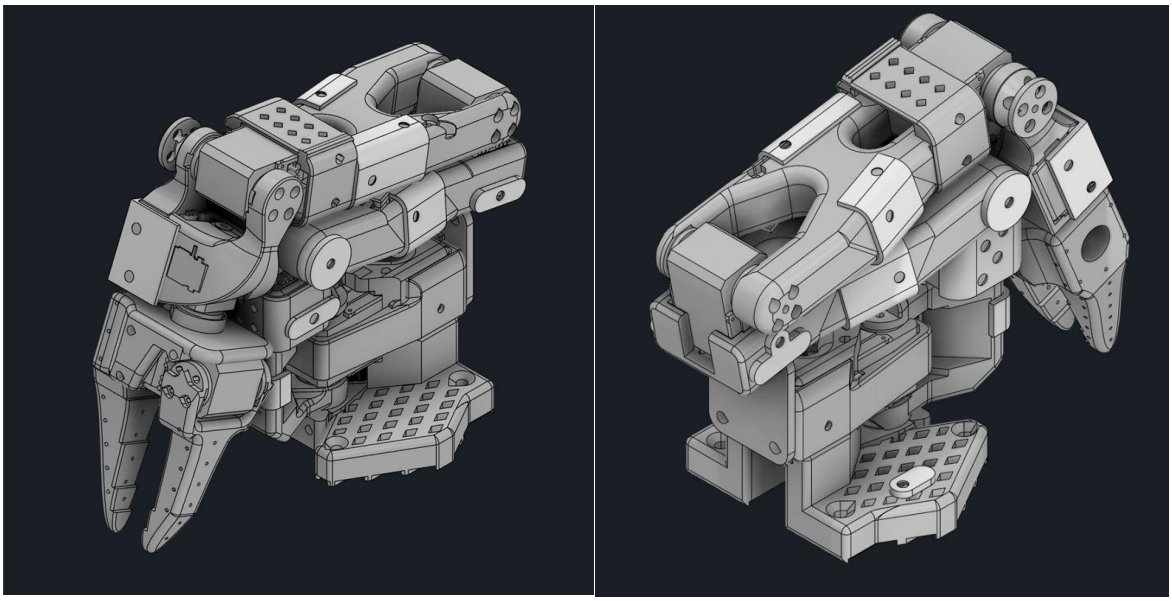


Figure 2: Placement of Tracker Ball Clips on SO-100 Robotic Follower Arm

To better facilitate accurate tracking of the links, circular clips were designed to index exactly at the rotational origin axis of each link. All tracked samples shared this same origin point to allow for accurate comparison across samples. Three other tracker balls were placed in a unique randomized pattern to allow for the mocap system to distinguish between bodies.

The arms were arranged in the mocap space such that each could be viewed by the optical tracking cameras during the same trial. With this setup, the position of each link was tracked, with the change in position relative to the starting position of each link being the point of interest. With two arms in the setup at a time, the required number of tracking balls were:

$$2 \text{ arms} \times 5 \text{ links/arm} \times 4 \text{ tracking balls/Link} = 40 \text{ tracking balls}$$

Figure 3 shows a simplified illustration of this experimental setup, with a leader and a follower arm within range of the Vicon cameras such that their positions could be recorded. It is important to note that for each trial, the arms were placed in a position at the limit of physical movement to allow for “homing” of the system; this was done despite the usage of servo motors in the robot arms to account for any physical deviance. The arms then began teleoperation, where a human operator moved the leader arm to various static positions throughout the entire range of motion of the system. A stop of 5 seconds was made at each static hold position.

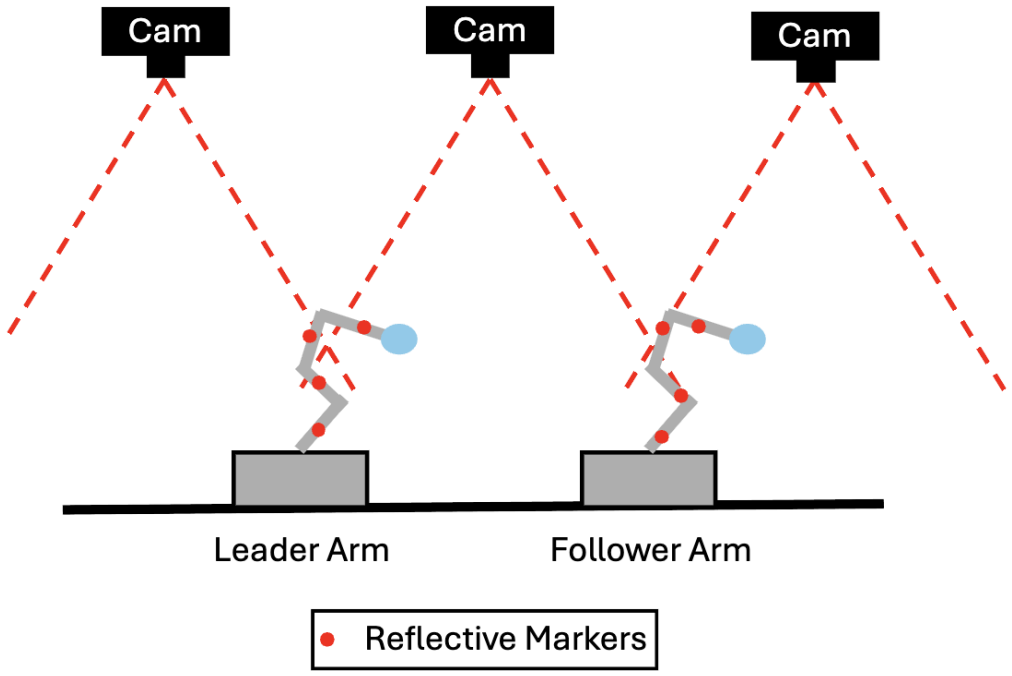


Figure 3: Simplified Experimental Setup of SO-100 Arms Tracked with Mocap System

The experiment was then modified; with the consideration that only the end effector of the robot arm would interact with any use case, tracking clips were removed for all joints except for the end effector to allow for more detailed analysis of the position deviation. By reducing the number of tracking points, overall motion capture fidelity and reliability were increased significantly, allowing for the more accurate capture of data. The same measurement was then taken with this new setup, teleoperating the robot arms throughout the entire range of motion with static holds made at various positions.

Measurement Chain

The measurement chain followed the control path from human input to robotic output and how the data was obtained from the Vicon Tracker. A human operator moved the leader SO-100 arm through an arbitrary trajectory. The leader arm encoded this motion and sent it via USB to a computer. The encoder data of the leader arm was sent to the follower arm. The data was decoded by the follower arm's onboard MCU, and the position data was then written to the motors to execute final movement. The data obtained by the motion capture system was the position and attitude of each link of interest. After measurements were completed, this data was then used to calculate the position and

orientation of the links relative to the base of the arms. Figure 4 represents an illustration of the signal chain in the system. Due to the digital nature of the signal chain, error checking ensures full signal integrity; any difference between the leader's position and the follower's position came from mechanical compliance, servo motor characteristics, assembly differences, or other physical limitations of the low-cost platform.

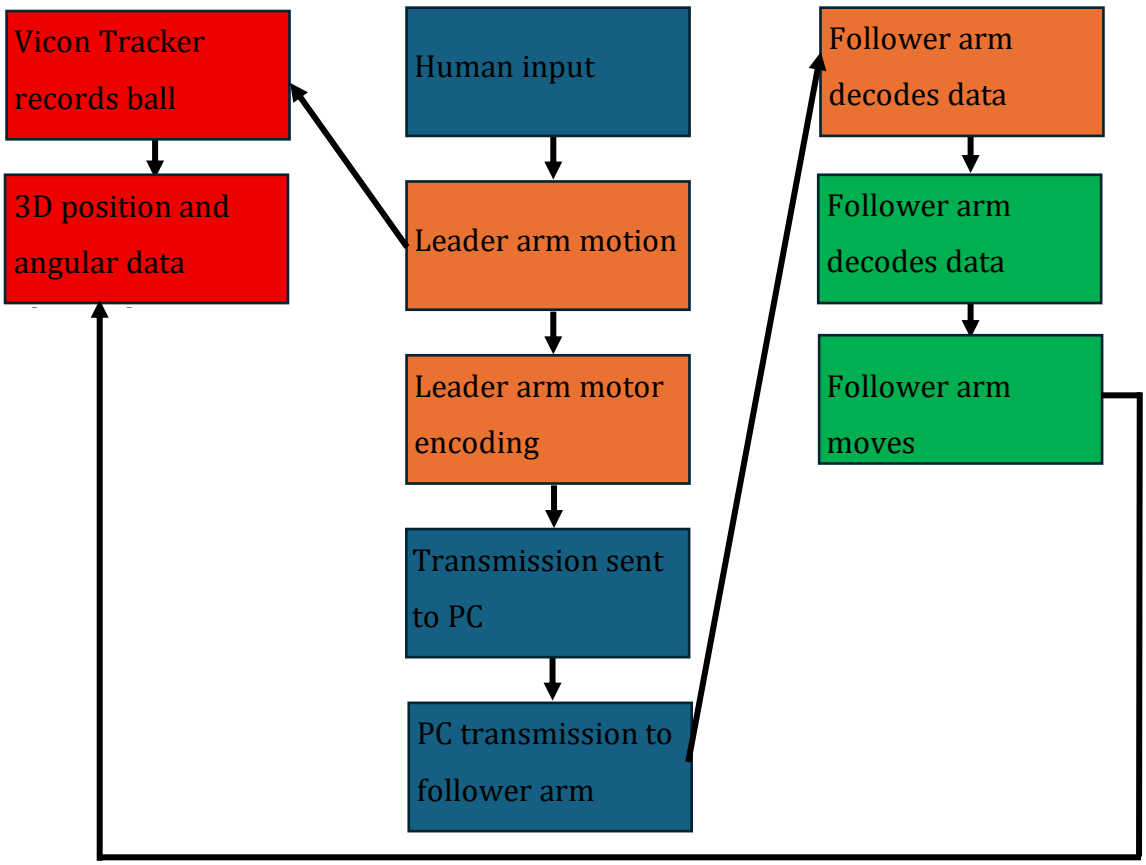


Figure 4: Measurement Chain of Data Gathering Process

The main performance metrics were:

- Trajectory deviation**:** differences between each follower arm's physical trajectory and the master arm's physical trajectory.
- Unit-to-unit variability**:** differences in performance between the multiple samples of follower arms.
- Joint-level variation**:** differences between physical input and output across individual joints, which can be visualized using a heatmap.

Alternative Designs

A variety of alternative measurement systems were considered for the experiment. A proposed alternative design involved a three-dimensional repeatability test using a dial indicator and reference surface (such as a steel ball or other calibrated surface) to test for deviations in movement around or along the surface. This type of test is relatively common among high-precision machine tools; however, it was ultimately decided against due to the focus on analyzing the deviation between the input arm and output arm.

An additional proposed alternative measurement configuration involved a realistic use case of the SO-100 robot arms to measure the repeatability of the robot. A laser pointer would be affixed to the gripper of the arm, and the leader arm would point at a specific point relative to the leader origin. Through teleoperation, the same task would be replicated on the follower arm, then the deviation of the follower arm's laser location would be compared to determine the relative error through the signal chain.

Analysis Procedure

The Vicon motion-capture data was analyzed using Python to quantify how closely the follower robot arm matched the motion of the leader arm. Each trial contained position and attitude data for rigid bodies on the leader and follower arms. Although the Vicon files included both translation and rotation data, only the translation coordinates TX, TY, and TZ were used for the position-error analysis. These coordinates represented the 3D position of each tracked body in millimeters.

The first step was to assign a time value to each Vicon frame. The time vector was calculated from the Vicon frame number and the Vicon sampling frequency. Time was calculated as:

$$t_i = \frac{F_i - F_0}{f_s} \tag{1}$$

where t_i is the time at frame i, F_i is the Vicon frame number, F_0 is the first frame number in the trial, and f_s is the sampling frequency. The data was sorted by frame and subframe before calculating time and duplicate frame entries were removed so that each time point was only represented once.

The raw motion capture data contained multiple missing points, sudden jumps, and unrealistic values well beyond the physical limits. These were caused by any occlusion, reflection, or other errors by the motion capture system. Values that exceeded physical limits or were otherwise unreasonable were removed before performing data analysis. Each tracked body position was represented as

$$p_i = \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} \tag{2}$$

where x_i, y_i and z_i are the TX, TY, and TZ coordinates at frame i.

The first cleaning step removed unrealistic frame-to-frame jumps. The change in position between adjacent frames was calculated as

$$\Delta p_i = p_i - p_{i-1} \tag{3}$$

and the corresponding 3D position speed was calculated as

$$v_i = \frac{\left|\left|\Delta p_i\right|\right|}{t_i - t_{i-1}} \tag{4}$$

where $\left|\left|\Delta p_i\right|\right|$ is the Euclidean distance between consecutive position measurements. Any frame-to-frame motion greater than 2000mm/s was treated as an unrealistic Vicon tracking jump and removed from the data.

The second step removed local spikes. A rolling median position was calculated using a 0.09s window and each point was compared to this local median. The spike distance was calculated as:

$$d_{spike} = \left|\left|p_i - \bar{p}_i\right|\right| \tag{5}$$

where $\bar{p}_i$ is the rolling median position at frame i. If a point was more than 50mm away from the rolling median, it was removed.

For the joint-tracking data, an additional link-distance consistency check was applied. This was used because the original goal was to track several joints, but some joints had inconsistent marker tracking. The distance between adjacent tracked bodies, such as

shoulder pan to shoulder lift, shoulder lift to elbow, and elbow to wrist, was calculated for each frame. The typical distance for each adjacent pair was estimated using the median link distance for that trial. Frames were rejected if the measured link distance differed from the median by more than 100 mm:

$$\left|\left|L_i - \bar{L}_i\right|\right| > 100mm \tag{6}$$

where L_i is the measured link distance at frame i and $\bar{L}_i$ is the median link distance for that adjacent body pair.

After cleaning the data, the raw position data (with regard to the global coordinate system) were converted to a relative position (with regard to each robot's base position). Thus, the analysis compared how each robot moved relative to its local origin:

$$\Delta p_i = p_i - p_0 \tag{7}$$

where p_i is the position at frame i and p_0 is the starting baseline position. This produced a delta trajectory for the leader and a delta trajectory for the follower. The follower-leader delta error was then calculated at each frame:

$$d_i = \Delta \mathrm{p}_{F,i} - \Delta \mathrm{p}_{\mathrm{L,i}}. \tag{8}$$

This gave separate error components in the x, y, z directions where:

$$d_i = \left[d_{x,i}, d_{y,i}, d_{z,i}\right]. \tag{9}$$

The total 3D error at each frame was calculated using the Euclidean distance:

$$e_i = \sqrt{d_{x,i}^2 + d_{y,i}^2 + d_{z,i}^2} \tag{10}$$

Stopped positions were found using the equation:

$$v_i = \frac{||p_i - p_{i-1}\,||}{t_i - t_{i-1}} \tag{11}$$

where $v_i = 0\ mm/s$ and e_i was determined at those stops.

Experimental Data

Tracking - All Joints

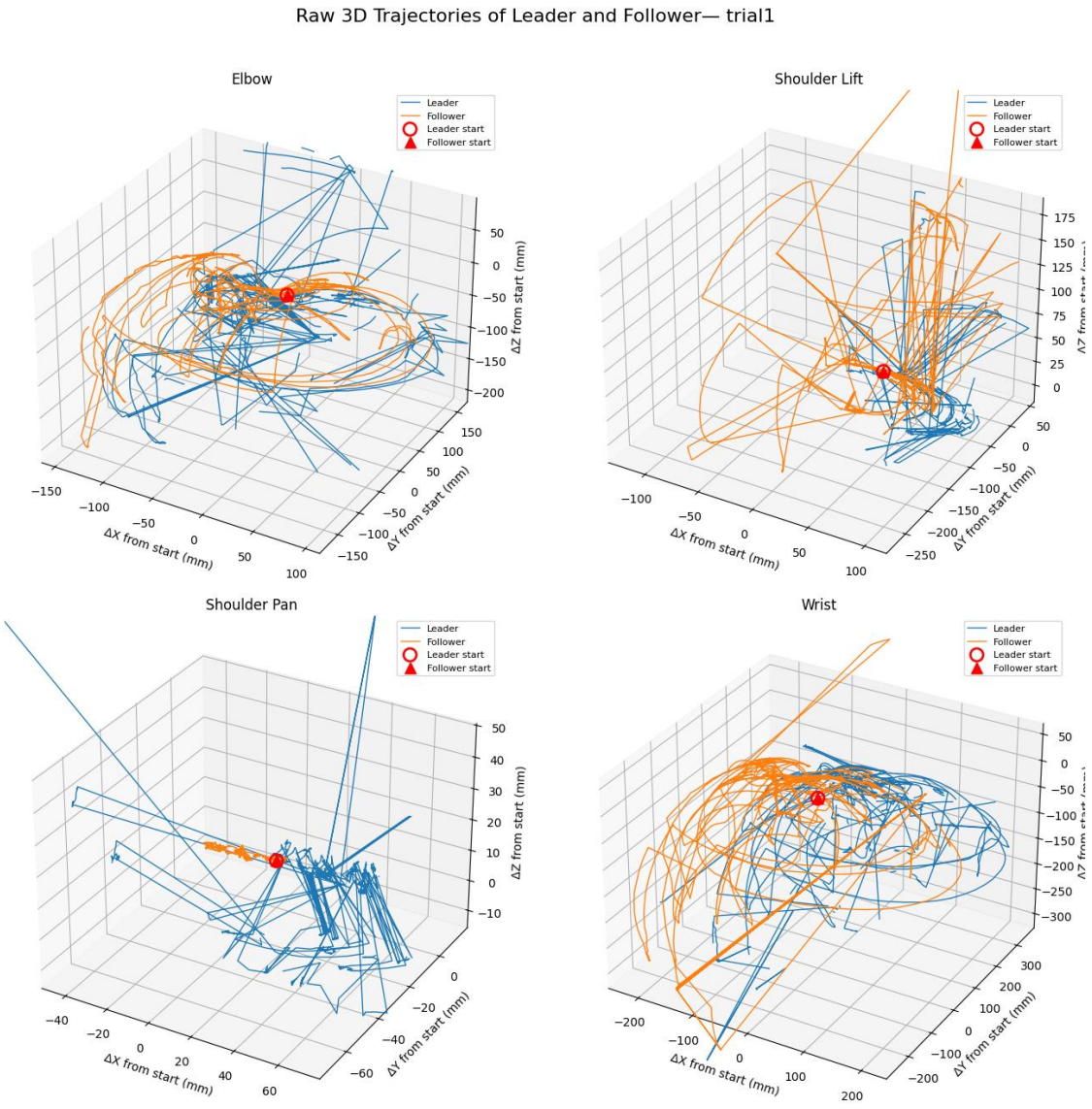


Figure 5: 3D Trajectories of Leader and Follower of All Joints – Trial 1

Figure 5 illustrates unprocessed positional plots of each link segment in Trial 1. The very large amount of deviation, spikes, jumps, flips and other errors well beyond physical reality are present.

Tracking - End Effector

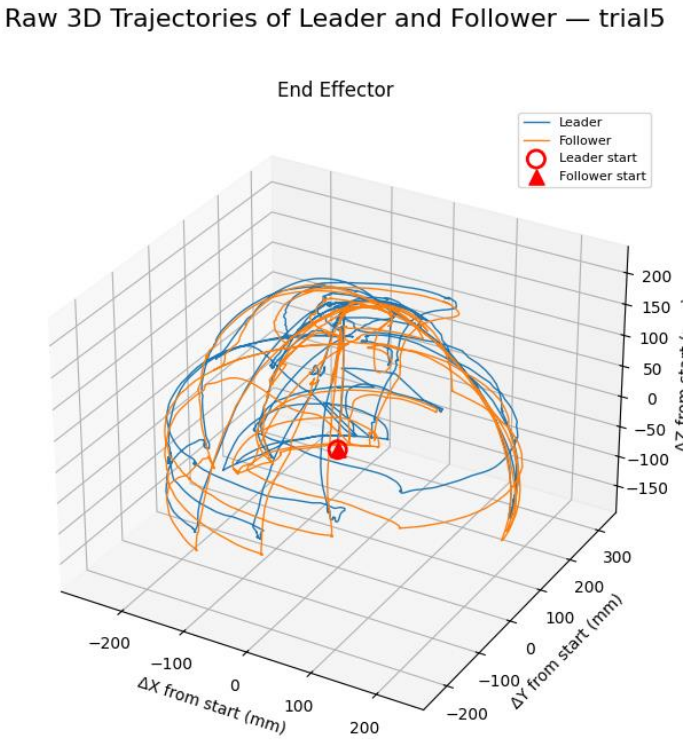


Figure 6: 3D trajectory of leader and follower end effector trial 5

Figure 6 illustrates a 3D plot of the raw recorded position data. Observed that the general shape of the leader arm movement is well replicated by the follower arm, the arms are moved throughout the entire range of motion, and that the arms are offset between each other due to how they are mounted on the platform.

Sources of Error

The motion capture system was also assumed to be properly calibrated and configured, however for best results a calibration routine utilizing a linear stage would have provided assurance of accuracy and precision in the tracking software.

It was found that the motion capture system used was generally insufficient for tracking large numbers of tracking points. A common issue in the earlier multi-body tracking configuration was the constant loss of entire bodies from the tracking window, and when bodies were being tracked, a constant issue with coordinate inversion (rigid bodies would appear as if they were inverted about their local origin point) was encountered. This in

particular was very detrimental to the data, as it caused extreme shifts in the recorded body position.

In the second test configuration involving only the end effector position (utilizing only 3 tracked bodies), results were significantly improved with regard to tracking reliability for better analysis.

Data Analysis

Tracking - All Joints

Follower-Leader Error

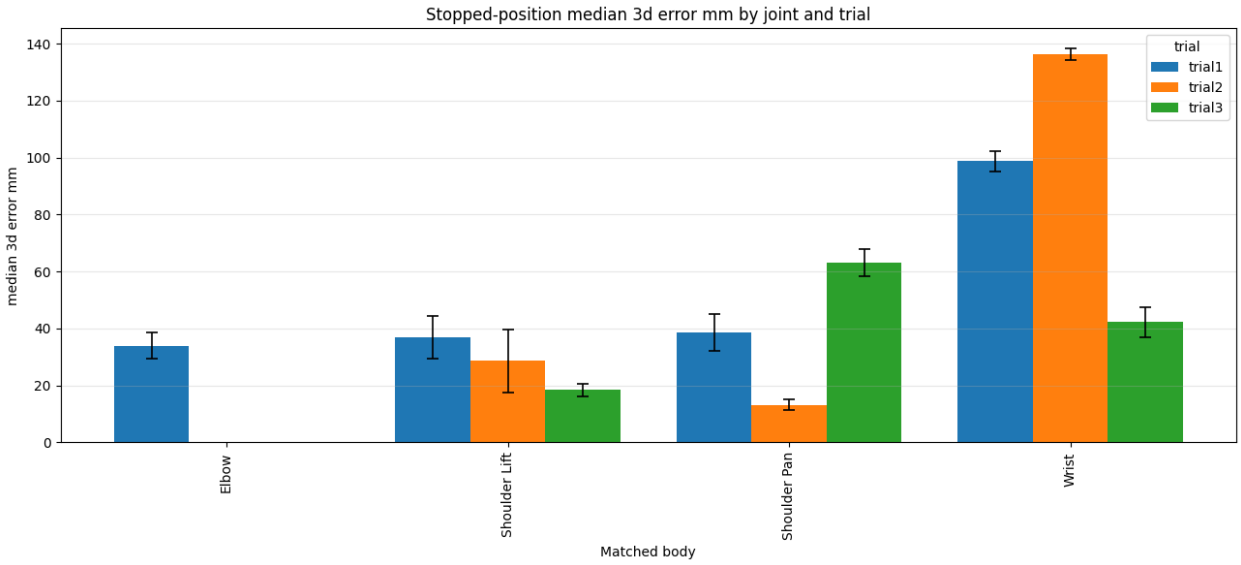


Figure 7: Stopped-position median 3D follower–leader error by joint and trial

In trials 2 and 3, elbow deviation data was discarded due to physically impossible amounts of deviation recorded in the data. Across all trials a general trend of increasing amounts of median error is observed as the joint move towards the end of the system, which aligns with expectations of compounding angular compliance.

Tracking - End Effector

Follower-Leader Error

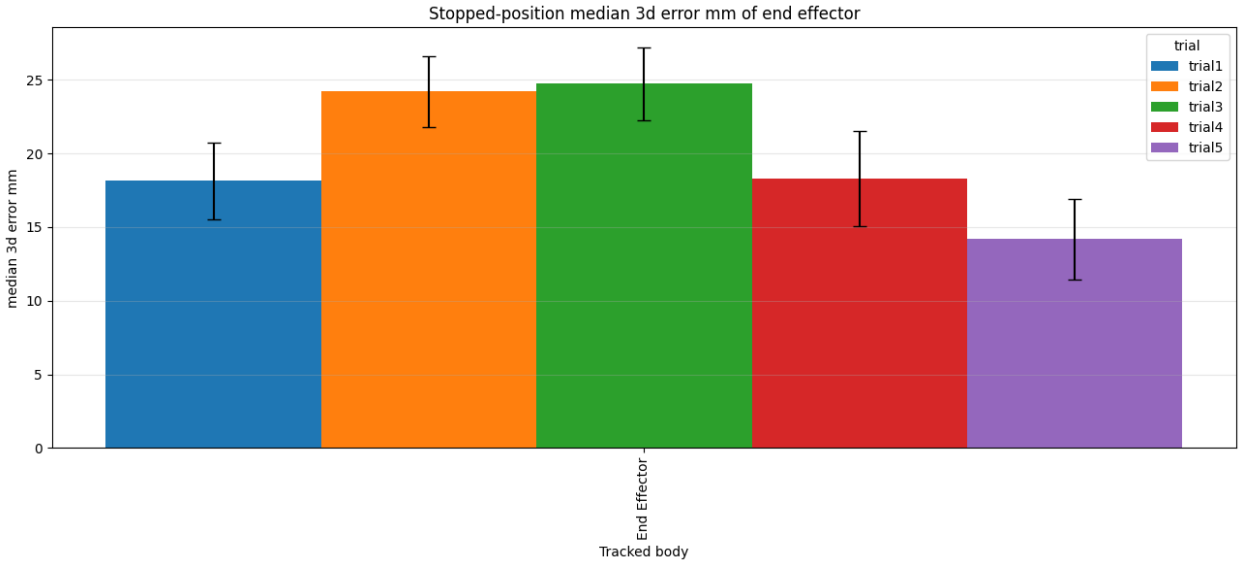


Figure 8: Stopped-position median 3D follower–leader error of end effector

Overall 3D Deviation between Follower and Leader

# of static positions	Mean (mm)	Median (mm)	Standard Deviation (mm)	Minimum (mm)	Maximum (mm)
88	21.895 $\pm$ 1.256	21.132 $\pm$ 1.723	11.963 $\pm$ 1.143	2.125 $\pm$ 0.529	65.401 $\pm$ 7.183

Table 1: 3D follower–leader error of end effector at static hold positions

The end effector-configuration positional deviance data reveals the final metric of interest. A median deviation for each trial run is shown in Figure 8. Table 1 lists the overall data summary of the end effector configuration tests. A nominal average of 21.895mm deviation between the input (leader) arm and output (follower) arm was determined, indicating that a relatively high degree of deviation is present.

Conclusion

Summary

This experiment quantified the trajectory deviation, precision, and repeatability of the low-cost SO-100 robotic arm platform. A Vicon optical motion-capture system tracked the physical motion of a teleoperated leader arm and a follower arm. The initial experimental design tracked multiple joints, but this caused multiple problems with reliability of the motion capture system.

The collected data revealed a trend of compounding angular compliance, with median error increasing as the joints moved closer to the end effector. The experimental setup was later simplified to track only the end effector, which significantly improved tracking reliability for analysis. After careful analysis of the data, an average 3-D positional deviation of $21.895 \pm 1.256mm$ was determined at the end effector, revealing that the SO-100 platform is ill suited for traditional high-precision use cases for robot arms, such as pick-and-place.

Recommendations

Users operating the SO-100 robotic arm should anticipate noticeable positional deviation at the end effector due to the compounding angular compliance of the joints. Because the platform utilizes 3D-printed components and low-cost servos, users should account for mechanical compliance and assembly differences when executing tasks that require precise motion reproduction. Particularly, while positional errors are not so high as to warrant labeling the device as a toy, it is ill suited for any realistic industrial or professional application.

A common application of such robot arms is to automate 3D printing systems by either removing finished prints or entire bed plates; a 2 cm positional deviation indicates tasks like accurately moving to grab a steel bed sheet would be exceedingly difficult. Commercially available videos of the SO-100 in action commonly demonstrate that in this sort of application, users resort to mechanically sweeping the bed with the arm to remove parts to get around this limitation.

Next Steps

Future testing should implement a calibration routine utilizing a linear stage to ensure the motion capture system is highly accurate and precise. Further experimental runs should strictly utilize the simplified end-effector tracking configuration to maintain data integrity and prevent the loss of body tracking. Additional analysis should also be conducted to isolate whether the physical limitations, such as servo behavior or the compliance of the 3D-printed links, are the primary contributors to the follower arm's deviation.

Appendix

3D Trajectories of Each Joint on Leader and Follower

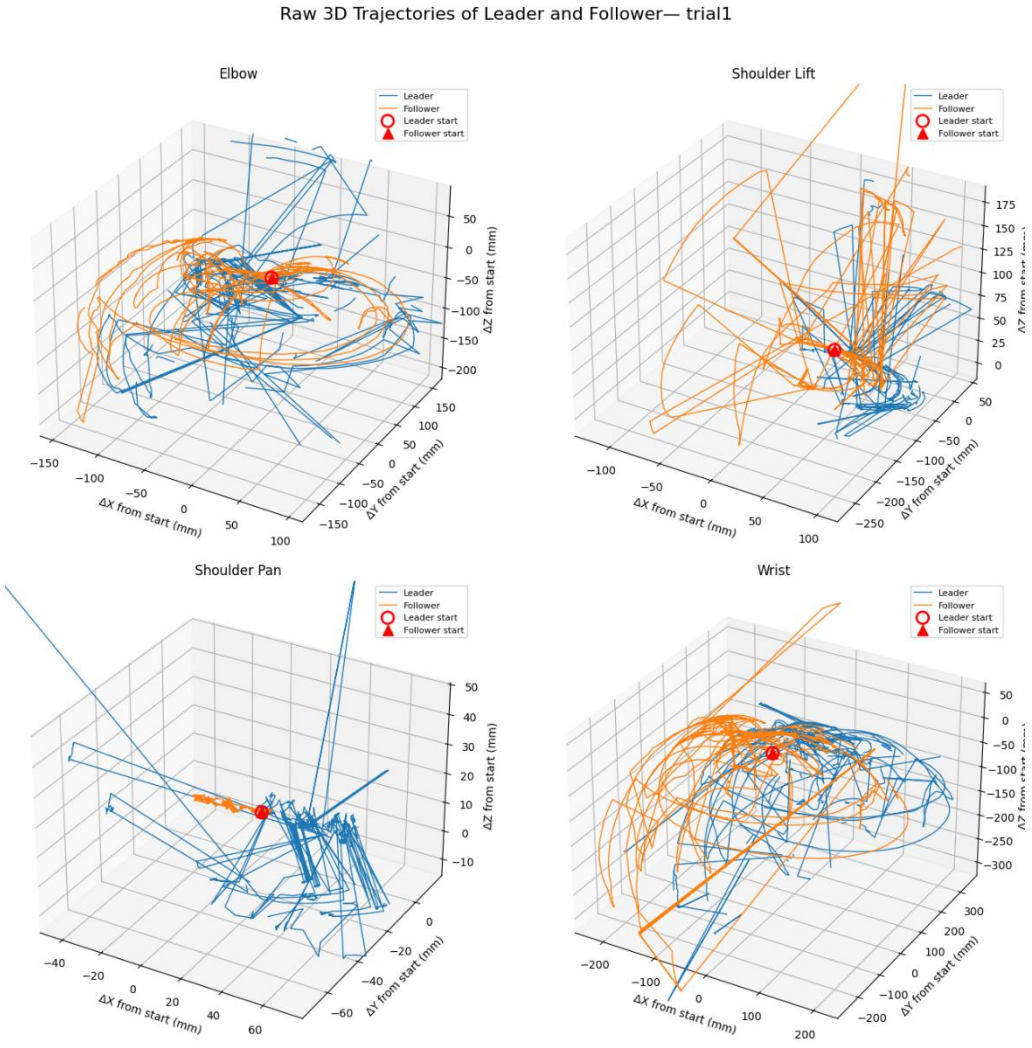


Figure 9: Trajectories of Each Joint of Leader and Follower - Trial 1

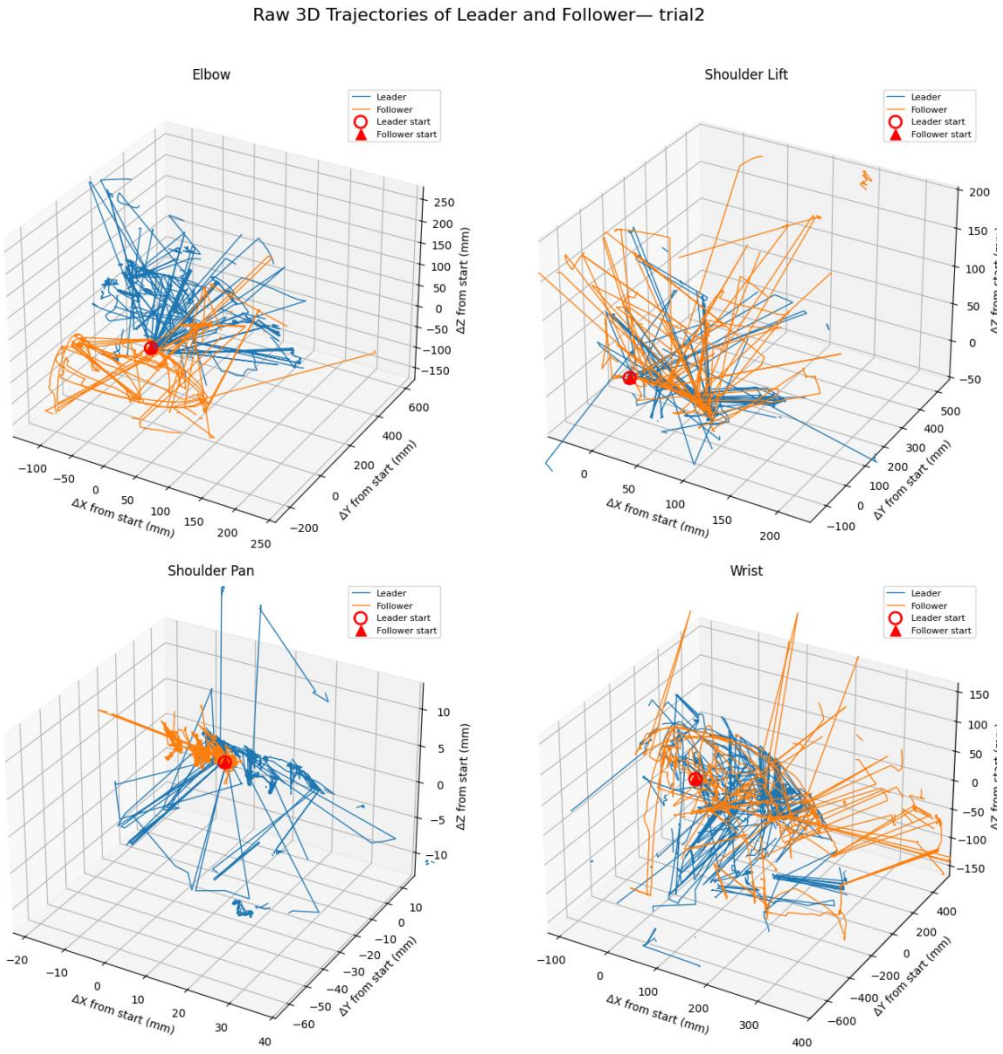


Figure 10: Trajectories of Each Joint of Leader and Follower - Trial 2

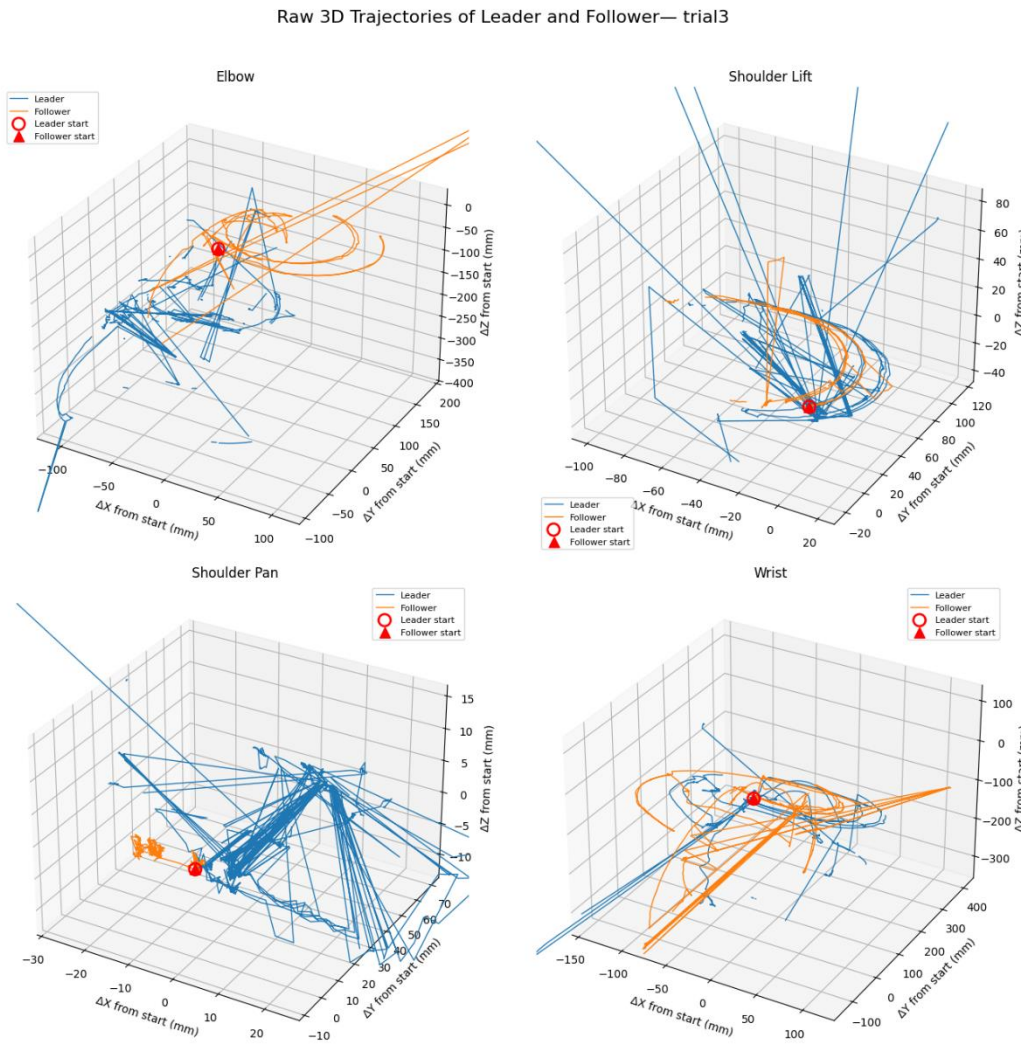


Figure 11: Trajectories of Each Joint of Leader and Follower - Trial 3

3D Trajectories of End Effector on Leader and Follower

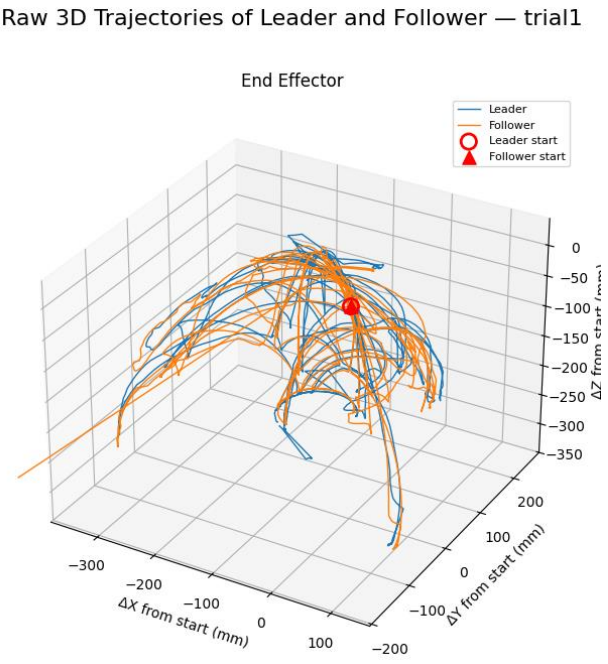


Figure 12: 3D Trajectories of Leader and Follower of All Joints – Trial 1

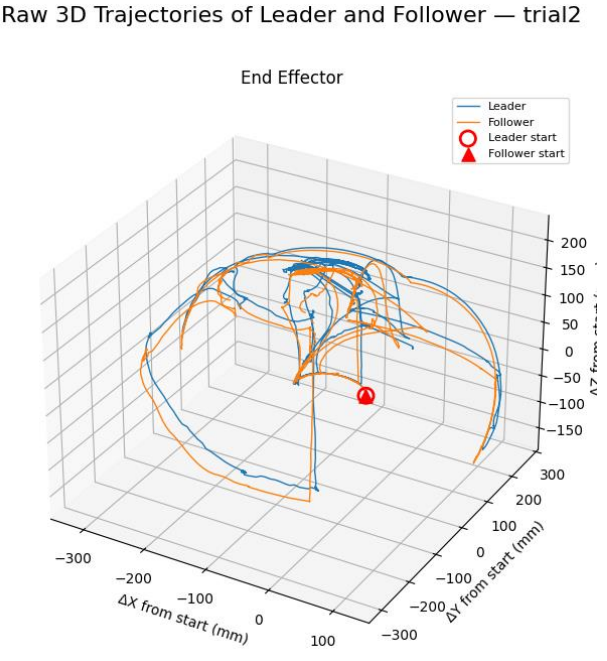


Figure 13: 3D Trajectories of Leader and Follower of All Joints – Trial 2

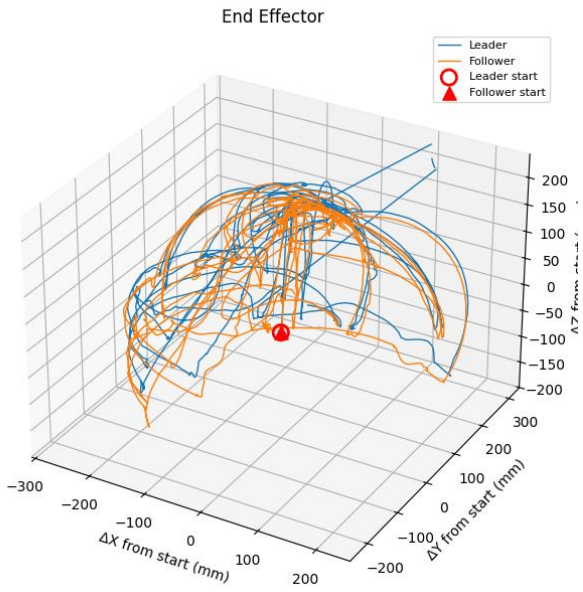


Figure 14: 3D Trajectories of Leader and Follower of All Joints – Trial 3

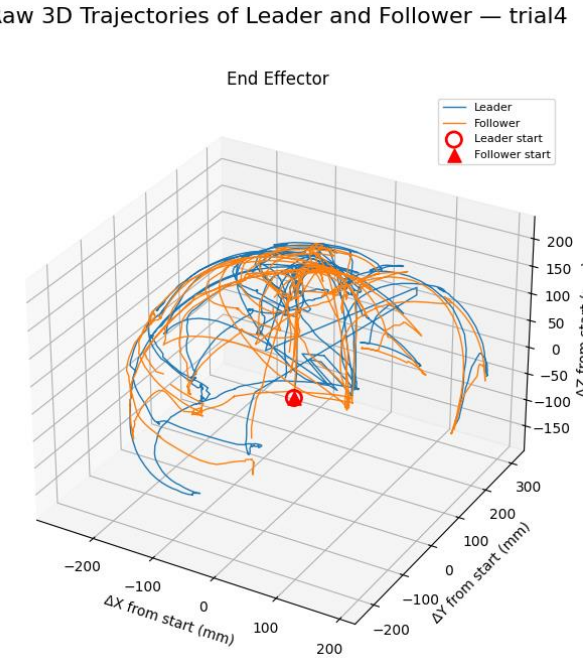


Figure 15: 3D Trajectories of Leader and Follower of All Joints – Trial 4

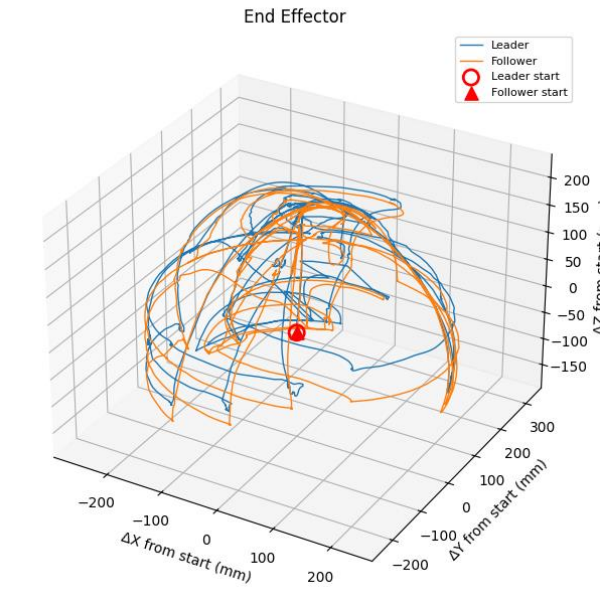


Figure 16: 3D Trajectories of Leader and Follower of All Joints – Trial 5

Follower Leader End Effector Error per Stop

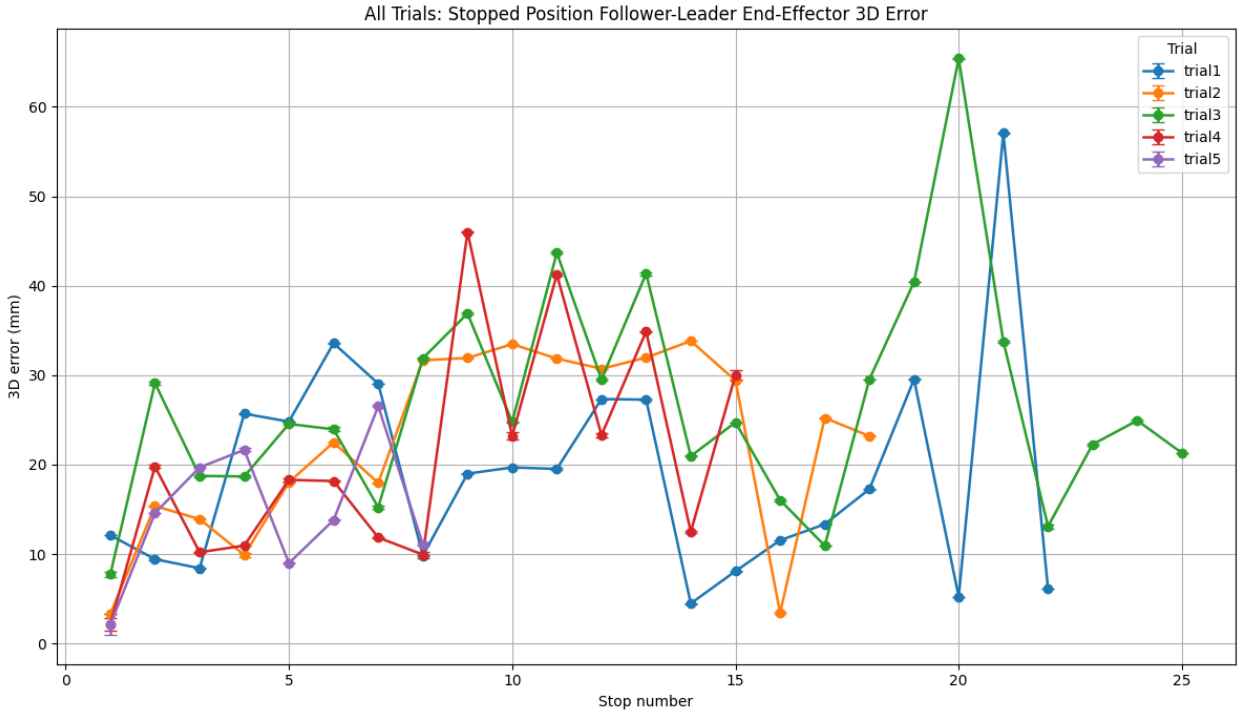


Figure 17: End effector positions across static hold positions

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Vicon Motion System Ltd. (2023). *Vantage Documentation*. Retrieved from Vicon Vantage Documentation: https://help.vicon.com/space/Vantage